Spreadsheets 1: structure, formatting, formulae and cell referencing

L11 FACT FILE

Specification: students should be able to

- Describe the basic structures of spreadsheet software: workbooks, worksheets, cells, rows, columns and ranges
- Describe and use data types; enter text, numbers and formulae; format cells, rows and columns
- Create and replicate formulae using arithmetic operators
- Use simple functions (SUM, AVERAGE, MAX, MIN, COUNT, COUNTIF) and relative and absolute cell referencing
- Improve worksheet presentation with headers and footers, and select areas of a spreadsheet for printing

CCEA fact file: Spreadsheets 1

Key words: worksheet, cell, row, column, cell reference, range, label, value, data format, formula, function, operator, replicate, fill handle, relative reference, absolute reference, merge and centre, print area



Spreadsheet formulas copied down: relative references change, the absolute reference stays fixed

How a spreadsheet is organised

A spreadsheet is application software that stores data in a grid so that it can be calculated, analysed and presented. One spreadsheet file, called a workbook, can contain several worksheets, each shown on its own tab. A worksheet is made of columns, labelled with letters, and rows, labelled with numbers. The box where a column and a row cross is a cell. Each cell has a cell reference, written column first and row second, so H8 means column H, row 8, just like co-ordinates in maths. A group of cells next to each other is a range, written with a colon between the first and last cell, for example B2:B10. Cells can hold labels (text that explains what data means), values (numbers used in calculations), dates, formulae, or a reference to another cell. The big advantage of a spreadsheet over a paper table is automatic recalculation: when one value changes, every formula that depends on it is worked out again straight away.

Data formats and presentation

Formatting changes how data looks without changing the value stored. A number can be shown as General, Number with a fixed number of decimal places, Currency (£), Percentage, Date or Time. Choosing the right format makes data easier to read and helps avoid mistakes: a cost column should be Currency, a discount Percentage, and a phone number Text so that the leading zero is kept. Percentage format shows 0.2 as 20%, so a rate typed as 20% is really stored as 0.2. Presentation features include bold or larger fonts for headings, borders, fill colour, alignment, merge and centre for a title that runs across several columns, and wrap text so long labels fit inside a cell. Column widths can be changed so that nothing shows as ####. For printing, headers and footers can add a title, file name, date or page number to every page, and a print area can be set so that only the part of the sheet that matters is printed.

Formulae and simple functions

A formula tells the spreadsheet what calculation to do. It must always begin with an equals sign, then use cell references, numbers and operators: + to add, - to subtract, * to multiply, / to divide and ^ for a power. Brackets are calculated first, so =(B2+C2)/2 finds the mean of two cells. Using cell references rather than typing the numbers means that the answer updates when the data changes. A function is a ready-made formula that does a particular job. SUM adds a range, AVERAGE finds the mean, MAX returns the largest value and MIN the smallest. COUNT counts how many cells in a range contain numbers, and COUNTIF counts only the cells that meet a condition, such as =COUNTIF(C2:C20,"Yes"). In each case the range goes inside the brackets with a colon between the first and last cell. =SUM(A1:A10) adds every cell from A1 to A10, not just the two cells named.

Replication and cell referencing

Replicating a formula means copying it into other cells, usually by dragging the fill handle down a column or across a row. This is quicker than typing each formula and makes errors less likely. By default references are relative: the spreadsheet adjusts them to fit the new position. If D2 contains =B2*C2 and it is filled down, D3 contains =B3*C3 and D9 contains =B9*C9. Sometimes every formula must point to one fixed cell, for example a single VAT rate, discount rate or exchange rate stored at the top of the sheet. This needs an absolute reference, made by putting a dollar sign in front of the column letter and the row number, such as F1. When =G2*(1+F1) is filled down, the G reference changes each row but F1 stays locked. If the dollar signs are missed, the copied formula points at the wrong cell, often a blank one, and gives wrong results without any warning. Keeping the rate in one cell also means it can be changed once and every result updates.

T Key terms from the mark schemes

Spreadsheet
Application software that organises data in a grid of rows and columns so it can be calculated, analysed and presented. A spreadsheet file (workbook) contains one or more worksheets.

Cell range
A block of adjacent cells referred to together, written first cell:last cell, e.g. B5:B12.
Credited: U1 2021 Q3, 2024 Q4, 2025 Q6

Label
Text used to identify the contents of a row or column, e.g. Name, Age.

Value
A number entered into a spreadsheet so that calculations can be carried out, e.g. 5, 23.

Formula
An instruction in a cell to carry out a calculation using operators and cell references, e.g. =A1+B1. It always starts with = and the cell shows the result, which updates when the referenced cells change.

Spreadsheet function
A built-in formula that does a specific job, e.g. SUM, AVERAGE, MAX, MIN, COUNT, IF and VLOOKUP.

SUM function
Adds all the numbers in a range, e.g. =SUM(A1:A10) adds A1 through to A10.
Credited: U1 2023 Q4, 2024 Q4, 2025 Q6

AVERAGE function
Calculates the arithmetic mean of the values in a range, e.g. =AVERAGE(A1:A10).

MAX function
Returns the highest value in a range, e.g. =MAX(A1:A10).

MIN function
Returns the lowest value in a range, e.g. =MIN(A1:A10).

COUNT / COUNTIF
COUNT counts the cells in a range that contain numbers; COUNTIF counts only the cells that meet a condition, e.g. =COUNTIF(C2:C20,"Yes").

Data types (spreadsheet formats)
Formats applied to cell contents to show what kind of data is stored, e.g. Currency, Number, Text, Date/Time, Percentage.

Replicating formulae
Copying a formula down or across cells (by dragging the fill handle or Fill Down) so the spreadsheet adjusts the references to match each row or column. Saves time and reduces errors.
Credited: Replication

Fill handle
The small square at the bottom right of a selected cell; dragging it replicates the cell's formula into neighbouring cells.

Relative cell reference
A cell reference that changes when the formula is replicated, so it refers to the matching cell in the new row or column, e.g. B9.
Credited: U1 2021 Q3

Absolute cell reference
A cell reference that stays the same when the formula is copied, replicated or dragged down, shown with $ signs, e.g. B9.
Credited: U1 2021 Q3, 2023 Q4, 2024 Q4, 2025 Q6

Merge and centre
Combining two or more adjacent cells into a single cell and centring the contents across it.
Credited: U1 2022 Q3, 2023 Q4

Wrap text
Formatting that makes long text continue onto extra lines inside the cell instead of spilling into the next cell.
Credited: U1 2024 Q4

Headers and footers
Text placed at the top (header) or bottom (footer) of every printed page, e.g. a title, file name, date or page number.

Print area
A selected range of cells that is the only part of the worksheet printed.

Exam focus

Spreadsheet questions appear on every Unit 1 paper, usually worth 10 to 13 marks and based on a screenshot of a sheet. You will often be asked to write a formula or function for a named cell: SUM with a correct range (2023 Q4, 2024 Q4(a), 2025 Q6(d)), AVERAGE or MAX (2021 Q3), or a subtraction (2024 Q4). Full marks need the equals sign, the right function name, brackets and a colon in the range, for example =SUM(B5:B12). Using x instead of * or listing cells one by one when a range was asked for loses marks. Absolute referencing is tested often (2019 Q8(c), 2021 Q3, 2023 Q4(c), 2024 Q4(e)): say that the reference does not change when the formula is replicated, and explain why it matters in that sheet, such as one exchange rate used by every row. Formatting questions ask you to name or describe a feature you can see, such as merge and centre (2022 Q3(b), 2023 Q4(a)), currency (2022 Q3(a)) or wrap text (2024 Q4). Name the feature precisely and say what it does to the cells.

Unit 1 2025 Q6	[10]	Spreadsheet features, ranges, SUM, IF with absolute reference, lookup functions
Unit 1 2024 Q4	[13]	SUM, subtraction, wrap text, absolute references, chart ranges and type
Unit 1 2023 Q4	[11]	Merge and centre, SUM, formulae, absolute references, IF, charts, template
Unit 1 2022 Q3	[12]	Currency, merge and centre, formulae, ranges, nested IF, VLOOKUP
Unit 1 2021 Q3(a)–(i)	[12]	Formatting, AVERAGE, replication, referencing, templates, validation, nested IF, MAX, ranges, macros

? Check yourself

1. Describe the difference between a row and a column in a spreadsheet. [2]
2. A sports club records match fees in cells B2 to B12. Write a function to add the values in B2 to B12. [1]
3. Write a function to find the highest value in cells C2 to C20. [1]

Answers: 1. A row runs horizontally / across the sheet and is identified by a number (1) / A column runs vertically / down the sheet and is identified by a letter (1) 2. =SUM(B2:B12) (1) – equals sign, function and correct range all needed 3. =MAX(C2:C20) (1)